

Orbital Collapse Architecture: Institutional Friction as the Regulator of Systemic Collapse

Evidence from the 2008 crisis and four further empirical domains

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Data and code: github.com/lnzainos/The-shadow-Node-Theory*

Abstract

When a dominant institution fails, its assets and market position are not destroyed — they are **absorbed** by an identifiable successor, and both the speed *and the shape* of that absorption are empirically regular. We model the post-failure dominance ratio $A(\tau) = \text{mass_absorber}(\tau) / \text{mass_collapsed_peak}$ as a power law, $A(\tau) = c \cdot \tau^{\Delta}$, on a clock τ that starts at functional extinction, and treat the exponent **Delta** as a collapse coordinate orthogonal to the satellization exponent b .

Applied to the six canonical absorptions of the 2008–2009 crisis (Lehman Brothers, Bear Stearns, Washington Mutual, Wachovia, Merrill Lynch, Chrysler), the framework recovers statistically significant power-law absorption in every case ($R^2 = 0.85\text{--}0.99$, all $p < 0.01$), reconstructed entirely from primary sources. The ordering of Delta tracks the **resolution channel**: regulator-brokered failures (WaMu via FDIC, Bear Stearns via the Fed) are absorbed near-instantly, while the single disorderly bankruptcy (Lehman) is the slowest.

Our central claim is sharper than speed: **institutional friction does not merely slow collapse — it regularizes its shape into a smooth power law**. We test this against four further domains with real data. The decisive contrast is the **frictionless** domain of crypto-assets: LUNA (May 2022) collapses as a *catastrophic cliff* (super-exponential, 5.6 orders of magnitude in 11 days), FTX/FTT as a *floor-arrested* power law, and EOS as *erratic fragmentation* — none of them the orderly power law that regulated failures exhibit. Astrophysical collapse (a solar flare, $R^2=0.975$; a tidal disruption event, $\sim t^{-5/3}$) and a biological succession (Delta→Omicron) confirm the generality of the form.

We organize these into a five-mode taxonomy governed by **friction × trigger × (floor/ceiling)**, unified by a **Principle of Least Friction** (collapse follows the minimum-friction path; the friction field's geometry selects the mode). The policy implication is direct: **orderly resolution machinery is what buys a power-law decay instead of a cliff** — and frictionless venues are structurally prone to catastrophic, not graceful, failure.

1. Introduction

The 2008–2009 crisis was, structurally, an episode of accelerated **concentration**: failing institutions did not vanish, their balance sheets and franchises were captured by identifiable successors. JPMorgan absorbed both Bear Stearns and Washington Mutual; Bank of America absorbed Merrill Lynch. This is precisely the dynamic too-big-to-fail regulation aims to constrain.

We ask two quantitative questions: *how fast* does post-failure absorption happen, and — more importantly — *what governs its shape*? We show that absorption follows a power law whose exponent and **regularity**

encode the institutional environment in which the failure was processed. The financial cases are one slice of a broader program (Shadow Node Theory, SNT) that finds the same structure across demography, epidemiology and astrophysics; we use that breadth here only to argue that what we observe in finance is an instance of a general law, and to isolate, by contrast, what regulation specifically contributes.

2. Model

A system carries two orthogonal coordinates:

- **Satellization $\mathbf{b}$:** $R(t) = m_{\text{hub}}(t)/m_{\text{node}}(t) = a \cdot t^{\mathbf{b}}$ — how dominance evolves while the coupled relationship runs.
- **Collapse Δ :** $A(\tau) = c \cdot \tau^{\Delta}$, with $A = m_{\text{absorber}}(\tau)/m_{\text{collapsed_peak}}$ and τ = time since functional extinction.

Δ is fit by OLS on log–log axes. $\mathbf{b} \perp \Delta$ is a falsifiable claim ($\text{corr}(\mathbf{b}, \Delta) \approx 0$ across systems with both measured): collapse can strike at any point of a satellization trajectory, so it is an orthogonal axis, not a terminal stage.

ACO criterion (definitional): a case qualifies as Orbital Collapse only if (1) the hub undergoes functional extinction and (2) its mass is absorbed by a specific, identifiable successor. Failures that dissolve without capture are excluded.

3. Data

The six 2008–2009 cases are reconstructed from primary records: SEC EDGAR (8-K, S-4, 10-K), Federal Reserve Flow of Funds, FDIC failure/acquisition records, SIGTARP, the U.S. Treasury TARP tracker, and the Valukas Lehman Examiner Report. Monthly resolution from the trigger date forward. No synthetic values. Cross-domain data: Yahoo Finance (crypto), CoV-Spectrum/LAPIS (variant genomics), NOAA SWPC GOES (solar X-ray), NASA IRSA/ZTF (tidal disruption photometry), Maddison 2023 (historical).

4. Results

4.1 The 2008 cohort: the resolution channel is legible in Δ

Every case yields a significant power-law absorption. Ranked by absorption speed (time to 90% absorption, in hours):

Hub → Absorber	Resolution channel	Δ	R^2	time→90%
Washington Mutual → JPMorgan	FDIC receivership (P&A)	+0.009	0.946	21 h
Bear Stearns → JPMorgan	Fed-brokered (Maiden Lane)	+0.043	0.926	626 h
Wachovia → Wells Fargo	open-bank, FDIC-adjacent	+0.153	0.892	4,140 h
Merrill Lynch → Bank of America	negotiated merger	+0.217	0.846	7,122 h
Chrysler → Fiat + US Treasury	government-managed (TARP)	+0.138	0.990	16,071 h

Lehman Brothers → Barclays + JPMorgan	disorderly bankruptcy	+0.246	0.892	30,681 h
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The two fastest absorptions are the two most actively brokered failures; the slowest is the one allowed to proceed as a disorderly bankruptcy. The span is ~1,460×. WaMu's 21 h matches the real FDIC weekend seizure (~48 h) in order of magnitude — a sanity check that the method tracks reality.

4.2 Friction regularizes the shape (not just the speed)

In all six cases — and across the high-friction historical cases (Rome, USSR; $R^2=0.77\text{--}0.99$) — absorption is a **smooth power law** ($R^2=0.85\text{--}0.99$). This is the regulated regime: institutional scaffolding channels the failure into an orderly, scale-free decay.

4.3 The frictionless counterexamples (the regulatory payoff)

What does collapse look like with **no** institutional friction? Crypto-assets provide the natural experiment (real data, Yahoo Finance / user corpus):

Case (friction = 0)	Trigger	Floor?	Mode	Signature
LUNA / Terra (May 2022)	abrupt	none	Catastrophic Cliff	super-exponential; 5.6 OOM in 11 days; exponential beats power law and still under-fits (returns accelerate)
FTX / FTT (Nov 2022)	abrupt	floor ~\$1	Floor-Arrested	sharp drop to a residual floor; power-law-like ($R^2=0.875$)
EOS (2018–)	gradual	none	Cracquelure	erratic fragmentation; power-law $R^2=0.10\text{--}0.70$

None of these is the orderly power law of regulated failure. **Without institutional friction, collapse is catastrophic, arrested, or erratic — never graceful.** This is the core regulatory message: regulation is what converts a cliff into a power law.

4.4 Cross-domain universality of the form

The smooth-power-law (regulated) mode also appears where the regulating "friction" is *physical*: a GOES M6.9 solar flare decays as a power law ($R^2=0.975$; radiative/conductive cooling); the tidal disruption event AT2019qiz (a star accreted by a black hole — the most literal collapse-with-absorption) decays as $\sim t^{-5/3}$ ($R^2=0.84$), with disk viscosity as the literal friction. A bounded biological succession (Delta→Omicron, South Africa) is a fast logistic sweep ($k=0.218/\text{day}$). The mechanism is not finance-specific.

5. Taxonomy and unifying principle

Five modes, governed by friction × trigger × (floor/ceiling):

Mode	Condition	Shape
Regulated Orbital Decay	high friction (any trigger)	smooth power law
Cracquelure Decay	friction≈0 + gradual	erratic fragmentation
Floor-Arrested	friction≈0 + abrupt + floor	power law to a residual floor

Catastrophic Cliff	friction $\approx$ 0 + abrupt + no floor	super-exponential
Logistic Sweep	bounded magnitude	S-curve

Principle of Least Friction. Collapse follows the trajectory that minimizes integrated friction — gradient flow on a stability landscape. The geometry of the friction field ($\times$ trigger $\times$ floor) selects which mode emerges. Names have established anchors: the *catastrophic cliff* is a fold catastrophe (Thom); *cracquelure* is desiccation cracking (stress relief along least-resistance paths). The stability-landscape picture connects SNT to catastrophe theory, the Waddington landscape, ecological resilience (Holling) and climate tipping points. See figures/fig_paisajes_colapso and figures/fig_catastrofe_cuspide (SVG + PNG).

6. Regulatory implications

1. **Orderly resolution buys power-law decay instead of a cliff.** Delta and the regularity (R^2) of absorption are an *ex post* diagnostic of whether resolution machinery (Title II, FDIC single-point-of-entry) delivered the rapid, predictable transfer it promises. Low-Delta, high- R^2 = the regime worked; cliffs or erratic fragmentation = it did not, or was absent.

2. **Frictionless venues are structurally cliff-prone.** The 2022 crypto collapses are not anomalies — they are what the model predicts for abrupt failure with no institutional friction and no value floor. This is a financial-stability argument for extending resolution-style friction (circuit breakers, disclosure, capital/withdrawal gates) to crypto and shadow-banking venues.

3. **A temporal dimension for too-big-to-fail.** Delta quantifies *how fast* concentration completes after a failure (JPMorgan absorbing Bear + WaMu), complementing static concentration metrics.

7. Falsifiability

- **RC-Delta1 (orthogonality):** refuted if $\text{corr}(b, \text{Delta}) \neq 0$ across systems with both coordinates.
 - **RC-Delta2 (least friction):** refuted if a realized collapse takes a higher-friction path when a lower-friction one was available.
 - **RC-Delta3 (absorption):** refuted if the absorber's mass does not grow post-absorption.
 - **RC-Delta4 (regularization):** refuted if frictionless collapses fit a power law as well as regulated ones.
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8. Limitations

This is a preliminary draft. Findings are **correlational**: the domains differ in more than friction (timescale, the meaning of "mass", microstructure). The crypto sample is small ($n \approx 3$) and LUNA/the solar flare are collapse-*shape* evidence, not ACO absorptions with a single absorber. The TDE exponent (-1.07) is shallower than the canonical $-5/3$ (g-band, imperfect host subtraction); the point is power-law regularity, not the exact value. "Time to 90%" is threshold-dependent, though the abrupt<gradual ordering is robust to 0.5/0.9/0.95. A full version will operationalize "friction along a path", expand cases per mode, and test orthogonality directly.

9. Conclusion

Institutional failure is a transfer of dominance to an identifiable successor at an empirically regular rate and shape. That shape — the collapse exponent Delta and its regularity — encodes the resolution regime. The decisive, policy-relevant finding is that **institutional friction is what makes collapse follow an orderly power law; remove it and failure becomes a catastrophic cliff**. For a conference on financial regulation in an era of disruption, this offers a compact, falsifiable instrument for asking whether our resolution machinery delivers orderly transfers — and a structural argument for why frictionless venues fail catastrophically rather than gracefully.

Appendix — data sources by case

- **Lehman Brothers:** Valukas A.R. (2010), Examiner Report, SDNY; Fed Flow of Funds 2008–2013; SEC EDGAR 10-K Barclays & JPMorgan.
- **Bear Stearns:** FRBNY (2008), Maiden Lane; SEC 8-K JPMorgan Mar 2008; Sorkin (2009), *Too Big To Fail*.
- **Washington Mutual:** FDIC (2008), Sep 25 release; JPMorgan 8-K Sep 2008; FDIC Failed Bank List.
- **Wachovia:** SEC 8-K Wells Fargo Oct 2008; FDIC statement Oct 3 2008; Wessel (2009), *In Fed We Trust*.
- **Merrill Lynch:** SEC S-4 BofA Dec 2008; BofA 10-K 2009; Lewis testimony to the FCIC (2010).
- **Chrysler:** SIGTARP (2012); Chrysler Group 10-K 2010; US Treasury TARP tracker; Rattner (2010), *Overhaul*.
- **Crypto:** Yahoo Finance (LUNA1-USD, FTT-USD); user corpus (EOS, ETH).
- **Astro:** NOAA SWPC GOES X-ray; NASA IRSA/ZTF (AT2019qiz).
- **Biology:** CoV-Spectrum/LAPIS open (GenBank), South Africa.

Full corpus, code and reproduction: github.com/Inzainos/The-shadow-Node-Theory